

Ayaan Hafeez

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SUMMARY

Computer Engineering student with production experience spanning AI systems, full-stack development, quality assurance, and requirements engineering. Shipped LLM pipelines, NL2SQL chatbots, and REST APIs at Elev8AI while working directly with customers to define project requirements and translate business workflows into technical designs.

EDUCATION

University of Alberta

Edmonton, AB

BSc in Computer Engineering Co-op Software Option

May 2028

- **Coursework:** Data Structures & Algorithms, Computer Organization & Assembly, Software Requirements, Digital Logic, Signals & Systems, Database Management, Operating Systems, Software Testing & Maintenance

TECHNICAL SKILLS

Languages: Python, Java, C/C++, Rust, Swift, JavaScript/TypeScript, Kotlin, Lua, SQL, Bash, ARM/MIPS, VHDL

Frameworks: FastAPI, Next.js, React, LangChain, ratatui, tokio, AppKit/Cocoa, SQLAlchemy, scikit-learn, NumPy, pandas

Tools: Git, Docker, PostgreSQL, SQLite, Redis, ChromaDB, Selenium, Ollama, Gemini/OpenAI/Anthropic APIs, AWS, Stripe, tmux, mosh

Hardware / Vision: OpenCV, ML Kit, AWS Rekognition, FPGA, Arduino, Raspberry Pi, ARM Cortex-M4

EXPERIENCE

AI Developer

May 2025 – Present

Elev8AI

Remote

- Architected an NL2SQL network-monitoring chatbot using LangChain, Ollama, ChromaDB, and PostgreSQL with RAG retrieval and session-aware context management, achieving 85% answer accuracy on internal benchmarks
- Led customer requirements discussions for AI and ERP projects, translating business processes into scoped requirements, workflows, data models, and implementation plans
- Designed ERP modules and performed functional QA across end-to-end business scenarios, validating workflows with customers before delivery
- Built a full-stack social media platform with a FastAPI backend, PostgreSQL/Redis, RBAC and JWT authentication, Stripe subscriptions, ML identity verification, and a React/TypeScript frontend deployed through Docker and nginx

PROJECTS

Bare Metal Raspberry Pi OS | C, ARM Assembly, Raspberry Pi

Apr – Jul 2026

- Built a bare-metal OS kernel for Raspberry Pi 3 in C and ARM Assembly across 6 subsystems: bootloader, UART/GPIO drivers, interrupt controller, preemptive scheduler, system calls, and MMU-backed virtual memory, with zero external library or OS dependencies
- Implemented a preemptive process scheduler and fork-based process isolation using ARM exception levels and a per-process page table, supporting concurrent execution across all 4 CPU cores

Vorssaint – Open-Source Contribution | Swift, Core Graphics, macOS

Aug 2026

- Fixed a display-recovery bug in a 5.5K-star macOS utility with 115K+ GitHub release downloads that could leave laptops without a usable screen after disconnecting an external monitor
- Implemented topology-aware recovery that excludes virtual displays, restores stranded physical displays, and adds regression tests for headless configurations; merged in PR #624

macwifi – Wi-Fi TUI for macOS | Rust, ratatui, tokio, CoreWLAN

Jun – Jul 2026

- Built a clean-room macOS port of Impala in Rust with WPA-Enterprise and hidden-network support; designed a Unix-socket daemon/client architecture integrating CoreWLAN, Keychain, LaunchAgent, and code signing
- Exposed Wi-Fi operations and speed tests through a CLI with streamed JSONL events, enabling custom connectivity frontends including a SketchyBar Wi-Fi popup

ssht – Persistent SSH Sessions | Rust, ratatui, tokio, SQLite, tmux, mosh

Jun – Aug 2026

- Built a Rust SSH workflow for resumable tmux sessions with recursive config parsing, concurrent remote probes, SQLite history, fuzzy search, and Mosh/local-echo modes

Terminal Mystery – Terminal-Based Murder Mystery | Lua, LÖVE

May – Sep 2026

- Built a narrative puzzle game around a virtual filesystem where players investigate rooms, inspect evidence, and correlate clues using progressively unlocked terminal commands such as cd, cat, and grep